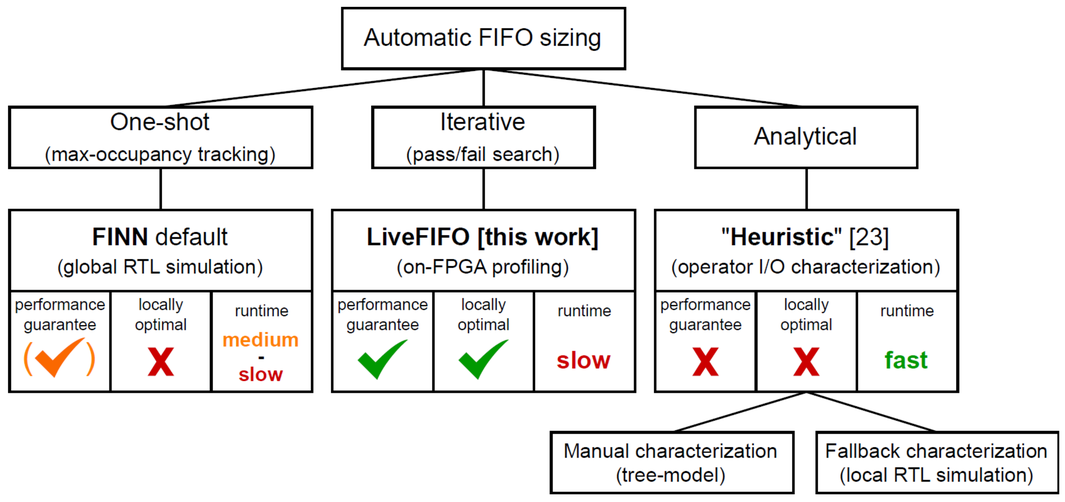
LiveFIFO:
FPGA-in-the-Loop Buffer Sizing for Dataflow Accelerators

Felix Jentzsch, Thomas B. Preußer, Lukas Stasytis, Yaman Umuroglu,
Jakoba Petri-Koenig, Christoph Berganski, and Marco Platzner

Background

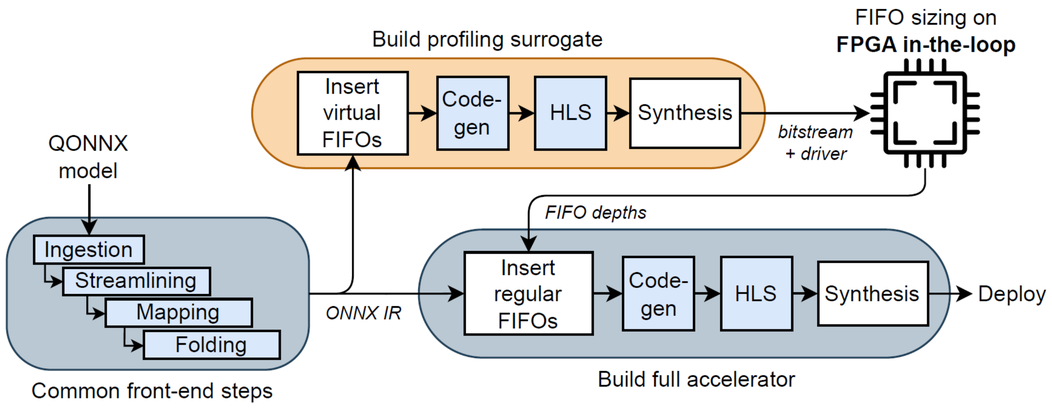
- **Motivation**
  - First-In-First-Out (FIFO) intermediate data buffers are crucial for dataflow accelerators
  - FIFOs prevent deadlocks and enable peak throughput by absorbing bursty I/O behavior of operators
  - Sizing FIFOs is a difficult problem
    - Too small: performance drop or deadlock
    - Too large: waste of resources
  - Multiple solutions already exist within FINN, but they are insufficient (see figure)

- **Our approach: Search-based FIFO sizing on a live FPGA ("in-the-loop")**
  - Synthesize a profiling surrogate design: prune data-path, insert "Virtual FIFOs"
  - Emulate FIFO depth configurations and measure their impact on accelerator performance
  - Running on FPGA gives us a 10.000x speedup over simulation, enabling iterative search (pass/fail)

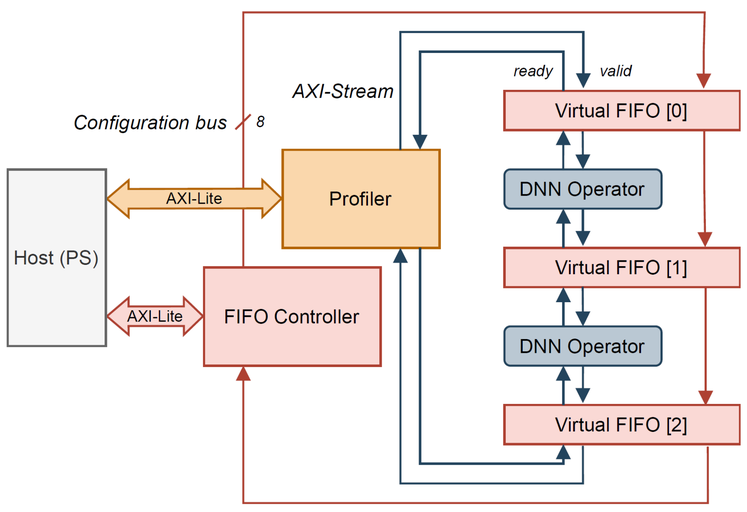


Taxonomy of FIFO sizing in FINN, categorized into one-shot approaches based on a single profiling run, iterative approaches performing multiple measurements, and analytical approaches relying on a-priori knowledge of operator behavior.
***[23] L. Stasytis**, F. Jentzsch, T. Preusser, Y. Umuroglu, J. Petri-Koenig, and Z. István,*
***„Heuristic & expert-guided buffer sizing for neural network inference applicationson FPGAs"**, FPT 2025*

Methodology

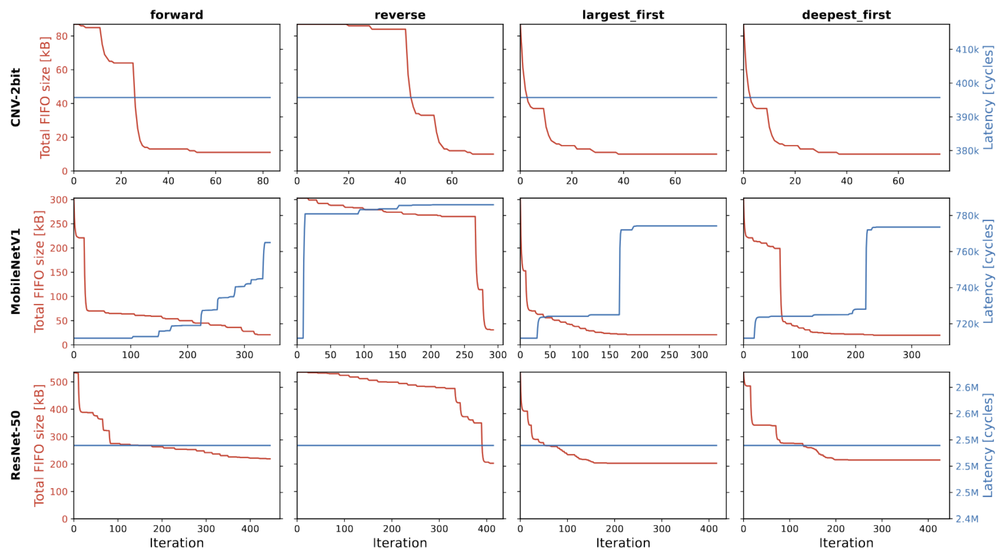


Simplified FINN tool flow incorporating FPGA-accelerated FIFO sizing.
Steps colored in blue can be re-used between both builds.



Simplified block diagram of the profiling surrogate design.
The AXI-Streamdata path is unconnected, leaving only the control flow to be implemented.

- **Three-phase FIFO sizing algorithm**
  - **Phase 1: Measure pipeline bottleneck interval.** Virtual FIFOs run in "detached" mode, operating as a free-running source/sink for adjacent operators and measuring frame initiation interval. Global bottleneck period determines the peak throughput of the entire dataflow pipeline.
  - **Phase 2: Measure maximum occupancy** as an upper bound for each FIFOs depth. Virtual FIFOs run in "paced" mode, operating like infinite FIFOs. Primary input is throttled to the bottleneck interval from phase 1.
  - **Phase 3: Binary search** starting from upper bound until stop condition (= performance degradation) is reached. This is repeated for each FIFO in a certain order.



Binary search phase for select models and search orders. The stop condition is set to interval degradation.
Only passing iterations are shown.

Results

Method	**LiveFIFO**		FINN		FINN		Heuristic		Heuristic	
Mode	best order		N=2		N=4†		rtlsim		tree-model	
Relaxation	no degradation		-		-		none		aggressive	
Model	Size [kB]	Intvl [cycles]	Size [rel.]	Intvl [degr.]	Size [rel.]	Intvl [degr.]	Size [rel.]	Intvl [degr.]	Size [rel.]	Intvl [degr.]
KWS	0.4	490	1.0x	62%	1.4x	1%	1.6x	0%	1.0x	0%
CyberSec	0.1	70	1.0x	0%	1.0x	0%	1.0x	0%	1.0x	0%
TFC-1bit	0.2	70	1.0x	10%	1.0x	0%	1.0x	0%	1.0x	0%
TFC-2bit	0.2	70	1.0x	10%	3.4x	0%	1.0x	0%	1.0x	0%
CNV-1bit	2.7	32774	1.8x	91%	1.8x	4%	1.2x	0.2%	0.7x	8%
CNV-2bit	10.1	115206	1.2x	14%	1.1x	13%	0.6x	0%	0.3x	21%
VGG10	2.2	1027	1.0x	0%	1.1x	0%	1.2x	0%	1.0x	0%
MobileNetV1	22.1	394272	13.0x	17%	33.6x	16%	8.6x	0.3%	4.9x	1%
ResNet-50	202.7	903174	2.7x	0%	4.8x	0%	18.9x	0%	6.9x	0%
RadioML T.	17.7	5120	1.3x	15%	1.7x	0%	–	–	–	–
Vision T.	85.0	81920	0.9x	60%	1.4x	0%	–	–	–	–
Language T.	84.5	81920	2.5x	–	4.6x	–	–	–	–	–

Comparison against baselines (FINN and Heuristic [23]), showing total FIFO size and interval degradation relative to our solution.
†Input throttling disabled.

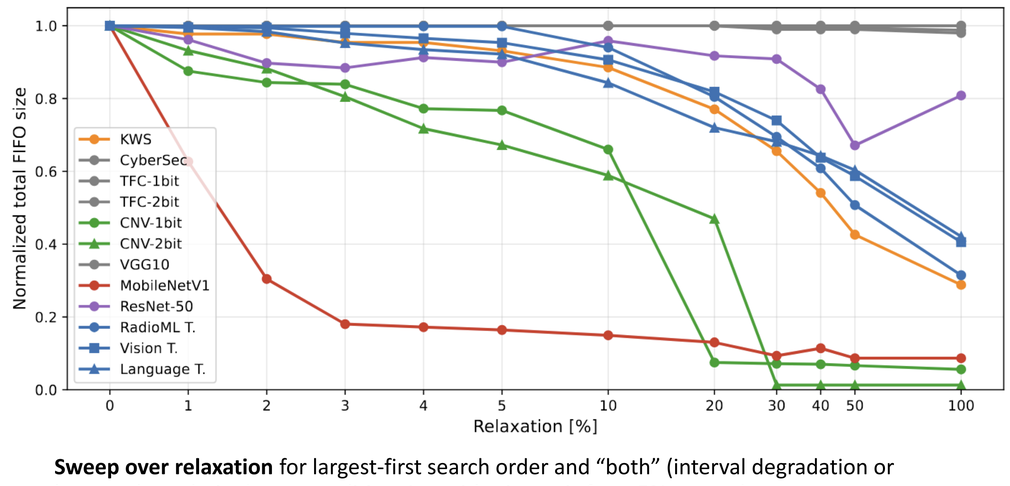
Method	Effective FIFO sizing time [s]				End-to-end build [min]	
	LiveFIFO	FINN	Heuristic	Heuristic	**LiveFIFO**	Heuristic
Mode	best order	N=4	rtlsim	tree-model	best order	tree-model
KWS	584 (1)	153	<1	<1	22	18
CyberSec	489 (1)	77	2	<1	18	13
TFC-1bit	576 (1)	113	<1	<1	20	16
TFC-2bit	573 (1)	88	<1	1	21	15
CNV-1bit	845 (1)	218	8	2	26	18
CNV-2bit	850 (2)	301	27	4	27	15
VGG10	1487 (1)	343	6	<1	52	29
MobileNetV1	2492 (16)	5008	470	110	80	47
ResNet-50	5140 (107)	19439	8104	324	307	227
RadioML T.	1964 (2)	279	–	–	75	–
Vision T.	2754 (10)	1314	–	–	79	–
Language T.	2768 (9)	1419	–	–	91	–

FIFO sizing runtime of our approach (including surrogate synthesis, search time on FPGA shown in brackets) compared to existing methods.

End-to-end build time is spent on generating a fully functional and FIFO-sized accelerator from scratch.

Performance Relaxation

- We usually stop at any interval or latency degradation (even 1 cycle)
- A user can relax the stop condition to trade-off performance against FIFO resource savings
- This might be attractive for certain models. Example: MobileNetV1, where a 3% relaxation reduces total FIFO size by >80%



Sweep over relaxation for largest-first search order and "both" (interval degradation or latency degradation) stop condition. Logarithmic x-axis from 5% onwards.

← ***Link to ARC 2026 paper***

PADERBORN UNIVERSITY

eki ENERGY EFFICIENT AI BY DNN APPROXIMATION FOR FPGAs

Supported by:
Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection
based on a decision of the German Bundestag

Contact: felix.jentzsch@upb.de | fpjentzsch | Felix Jentzsch | eki-project.tech | cs.uni-paderborn.de/ceg